

Lab 07 – Worksheet

Name: Group 07	ID:10303	Section: T-2
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b. Selective 2's complement

Inputs to complement				Outputs of the complement		
O	B ₂	B ₁	B ₀	C ₂	C ₁	C ₀
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	1	0	0
0	1	0	1	1	0	1
0	1	1	0	1	1	0
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	0	1	1	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	1
1	1	0	0	1	0	0
1	1	0	1	0	1	1
1	1	1	0	0	1	0
1	1	1	1	0	0	1

Write expressions for C_2 , C_1 and C_0

Table 7.2

$C_2 =$	$O'B_2 + B_2B_1'B_0' + OB_2'B_0 + OB_2'B_1$
$C_1 =$	$O'B_1 + B_1B_0 + OB_1B_0$
$C_0 =$	B_0

Provide appropriately commented code for your selective 2's complement module

```
module TwosCompliment(
    input O,
    input [0:2] B,
    output [0:2] C
);
    assign C[0] = B[2];
    assign C[1] = (~O & B[1]) | (B[1] & ~B[2]) | (O & ~B[1] & B[2]);
    assign C[2] = (~O & B[0]) | (B[0] & ~B[1] & ~B[2]) | (O & ~B[0] & B[2]) | (O & ~B[0]
    & B[1]);

endmodule
```

Write a testbench to thoroughly test designed complement module.

```
module comp_test();
    reg o;
    reg [0:2] b;
    wire [0:2] c;
    TwosCompliment (o, b, c);

    initial begin
        o=0; b=3'b000; #10
        o=0; b=3'b001; #10
        o=0; b=3'b010; #10
        o=0; b=3'b011; #10
        o=0; b=3'b100; #10
        o=0; b=3'b101; #10
        o=0; b=3'b110; #10
        o=0; b=3'b111; #10

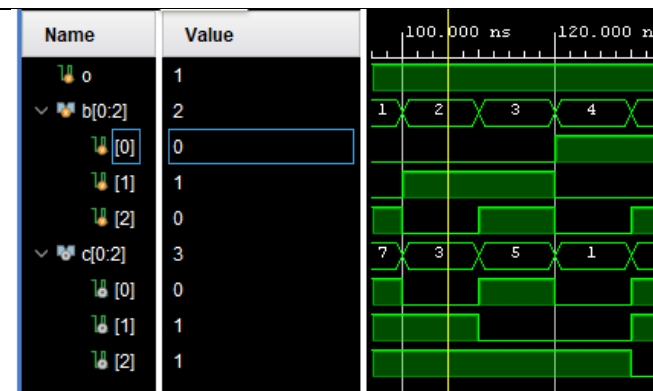
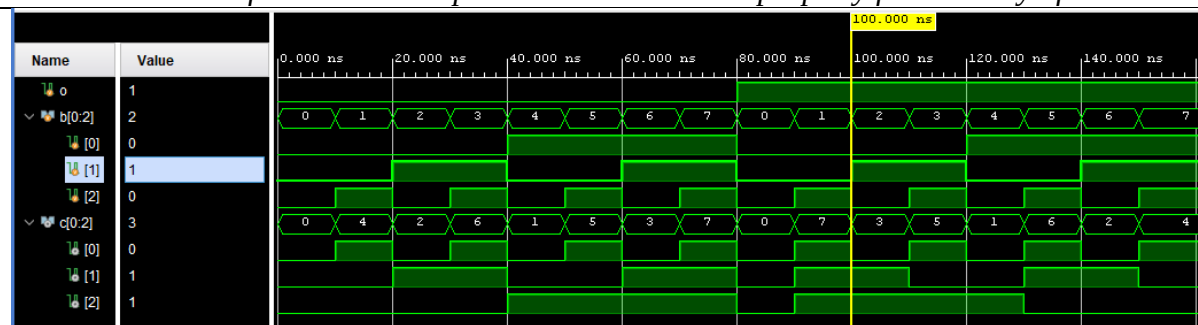
        o=1; b=3'b000; #10
        o=1; b=3'b001; #10
        o=1; b=3'b010; #10
        o=1; b=3'b011; #10
        o=1; b=3'b100; #10
        o=1; b=3'b101; #10
        o=1; b=3'b110; #10
    end
endmodule
```

```

o=1; b=3'b111;
end
endmodule

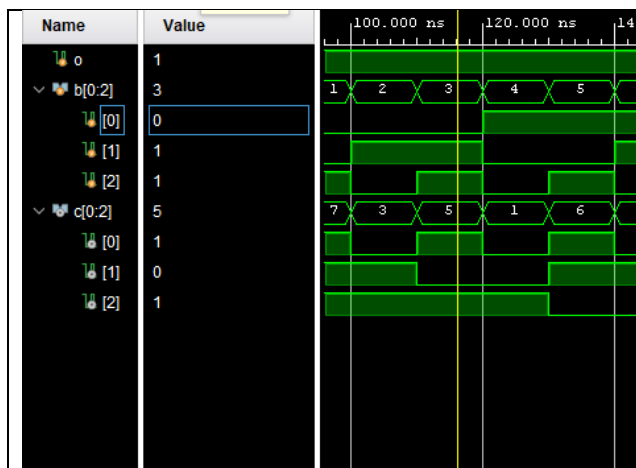
```

Attach screenshot of Simulation output- make sure to scale properly for visibility of all case.



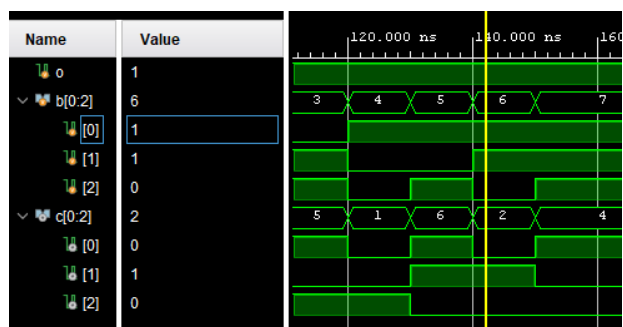
output = 010

2's complement of 010 is 011



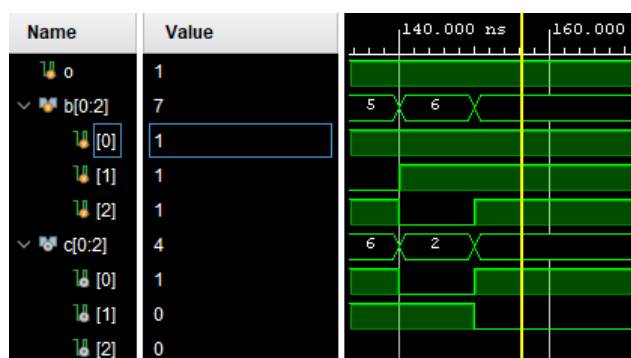
Output = 011

2's complement of 011 is 101



Output = 110

2's complement of 110 is 010



Output = 111

2's complement of 111 is 100

c. Adder/Subtractor system

Provide code for design module here

```

module adder_subtractor(
    input O,
    input [2:0] A,
    input [2:0] B,
    output [6:0] S,
    output Sign
);

wire [2:0] C;
wire [2:0] Su;
wire Cu;
wire sum;
wire [3:0] D;
wire AND;

complement gate1(O, B, C);
Threebitfulladder gate2(A, C, Su, Cu);

assign sum = (~O|Cu);
assign Sign = (~O|Cu);

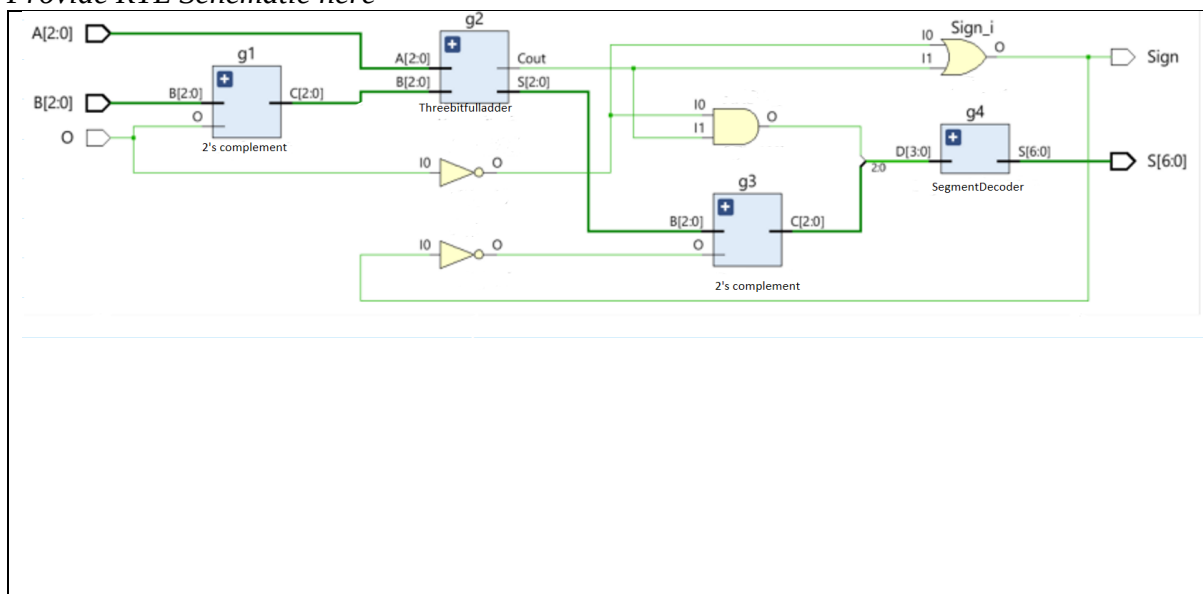
complement gate3(sb, Su, D[2:0]);
assign AND_1 = (~O & Cu);
assign D[3] = AND;

segmentDecoder gate4(D, S);

endmodule

```

Provide RTL Schematic here



Add your testbench here

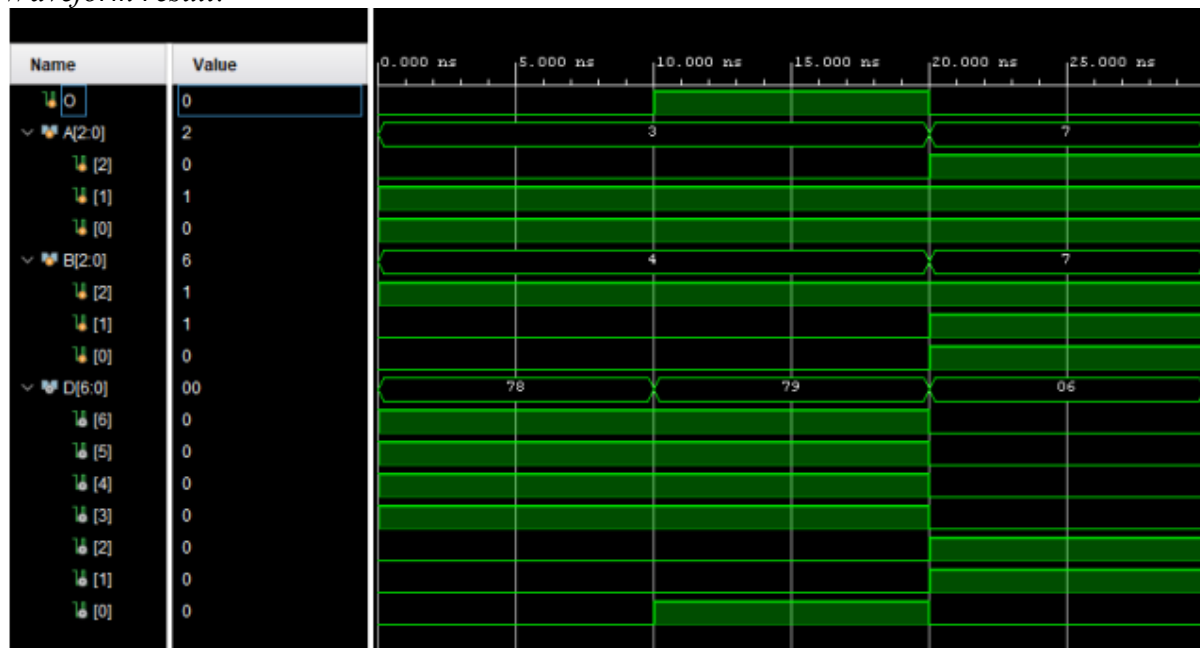
```
module adder_subtractor_testbench();

    reg O;
    reg [2:0] A;
    reg [2:0] B;
    wire [6:0] S;
    wire Sign;

    adder_subtractor t1(O, A, B, S, Sign);

    initial
    begin
        #100 O = 1'b0; B = 3'h3; A = 3'h4;
        #100 O = 1'b1; B = 3'h4; A = 3'h3;
        #100 O = 1'b0; B = 3'h7; A = 3'h7;
        #100 O = 1'b1; B = 3'h0; A = 3'h2;
    end
endmodule
```

Waveform result:



CASE 1:

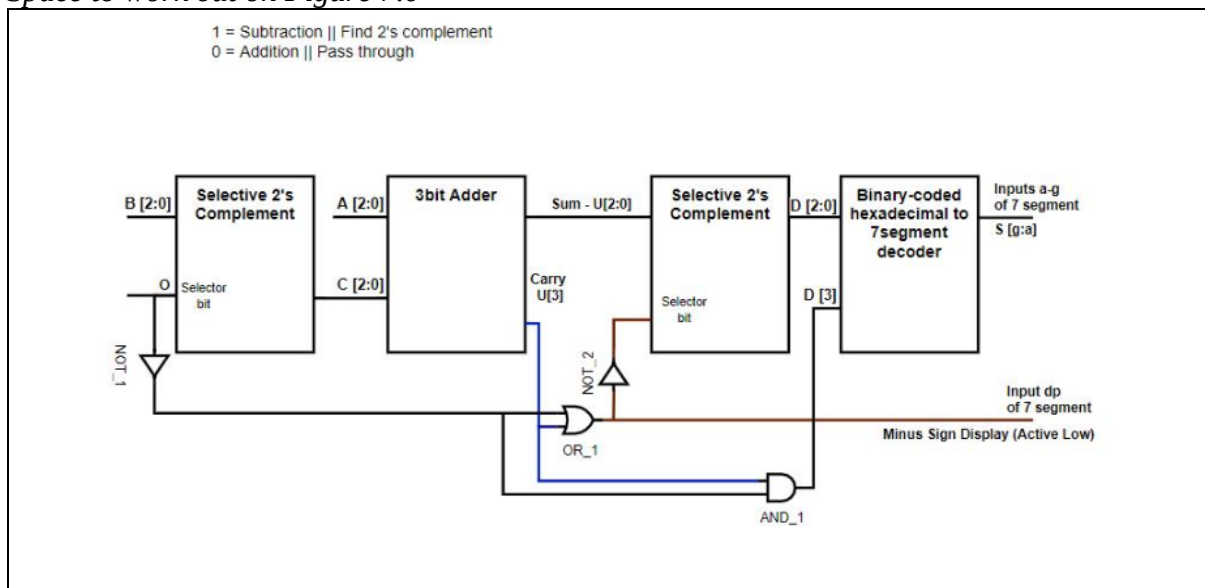
3-4 gives output as -1 which is correct

CASE 2:

7+7 segment gives output as "e" which is correct.

a. Exercise

Use Figure 6.0 and work out for cases when B is zero and subtracted from A. Reflect on the output obtained and suggest ways to fix (if required).

Space to work out on Figure 7.0*Space for reflection*

When analyzing the given circuit for the case where $B = 0$ and is subtracted from A , the expected output should simply be A itself, since subtracting zero does not alter a value. Since the subtraction operation involves finding the 2's complement of B , and B is zero in this case, its 2's complement remains zero. This means that the 3-bit adder is essentially adding $A + 0$, which should produce A as expected. However, the circuit also contains a second Selective 2's Complement block before the output is sent to the 7-segment display. If not controlled properly, this additional processing step might introduce an unintended transformation of the sum. A potential issue could arise if the system misinterprets the need for taking the 2's complement of the sum when it is not required, potentially flipping the sign representation of the result. If the circuit outputs A correctly but displays a negative sign, it suggests that $D[3]$ logic needs refinement. A fix would be to ensure that $D[3]$ is only activated when the result is genuinely negative rather than relying on the default subtraction logic.

Lab Evaluation Rubrics

Marks Distribution:

		LR2 Code	LR4 Data Collection	LR5 Results	LR7 Viva	LR10 Analysis
In-lab	Task b	20 points	5points	10 points	10 points	-
	Task c	20 points	-	20 points		-
Exercise	Task d	-	-	-		15 Points
Total Marks =	100	40	5	30	10	15

Marks Obtained:

		LR2 Code	LR4 Data Collection	LR5 Results	LR7 Viva	LR10 Analysis
In-lab	Task b					-
	Task c		-			-
Exercise	Task d	-	-	-		
Total Marks =	100					